

Zad 1

X ma rozkład $N(160.3, 9.8)$

$$P(151.5 \leq X < 168.5) = \Phi\left(\frac{168.5 - 160.3}{9.8}\right) - \Phi\left(\frac{151.5 - 160.3}{9.8}\right) = \\ = \Phi\left(\frac{8.2}{9.8}\right) - \Phi\left(\frac{-8.8}{9.8}\right) \approx \Phi(0.84) - \Phi(-0.90) = 0.79955 - (1 - 0.81594) = 0.61549 \approx 61.55\%$$

Zad 2

N - liczba pól

p - populacja łobioz

$$EX = 80 \cdot 0.54 = 43.2$$

$$D^2 X = 80 \cdot 0.54 \cdot (1 - 0.54) = 80 \cdot 0.54 \cdot 0.46 = 19.872$$

$$DX = \sqrt{19.872} \approx 4.46$$

$$N(43.2, 4.46)$$

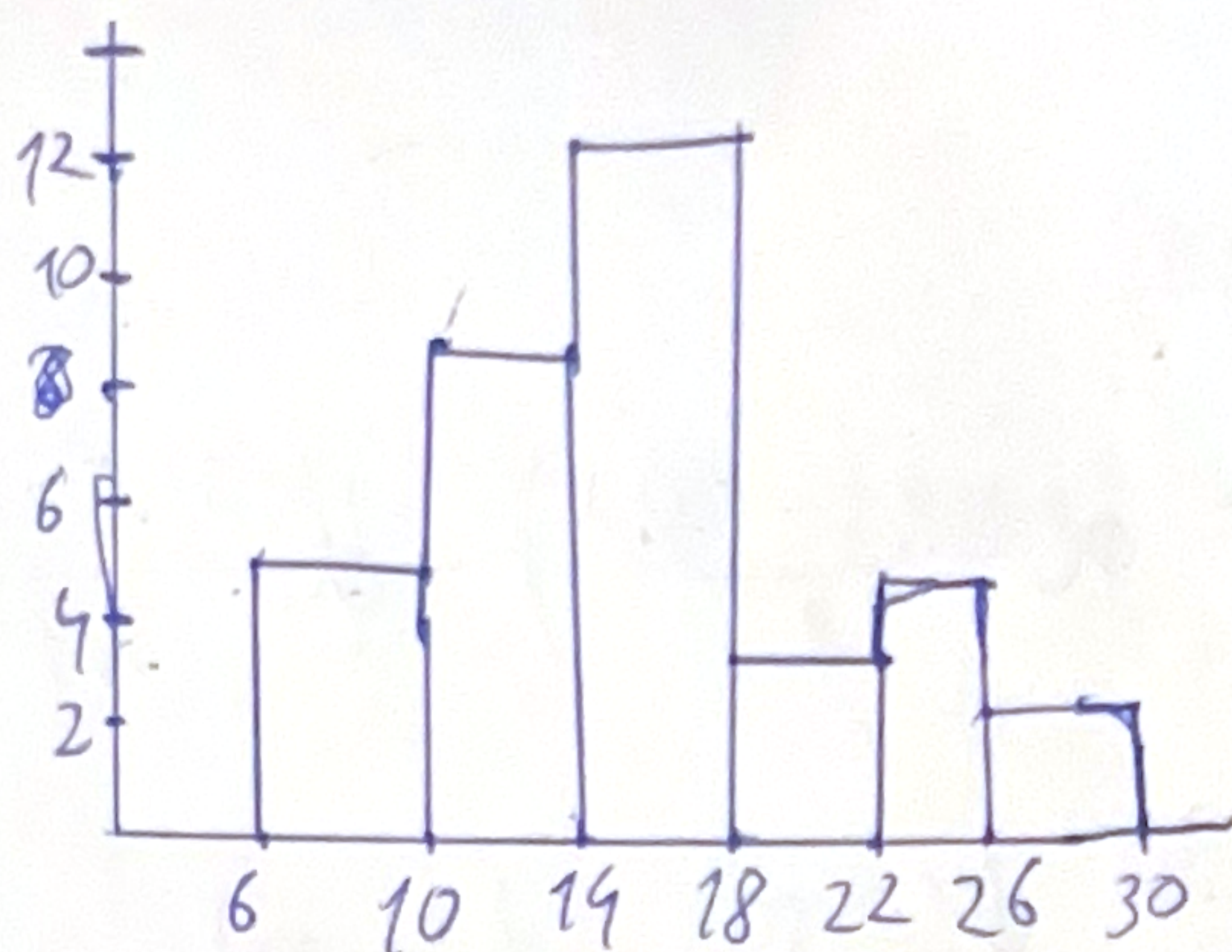
$$a) P(X > 50.5) = 1 - \Phi\left(\frac{50.5 - 43.2}{4.46}\right) = 1 - \Phi\left(\frac{7.3}{4.46}\right) \approx 1 - \Phi(1.64) = 1 - 0.9495 = 0.0505 = 5.05\%$$

$$b) P(X < 40.5) = \Phi\left(\frac{40.5 - 43.2}{4.46}\right) = \Phi\left(\frac{-2.7}{4.46}\right) = \Phi(-0.61) = 1 - 0.72907 = 0.27093 \approx 27.09\%$$



Zad 3

a)



$$b) \bar{x} = \frac{1}{35} (5 \cdot 8 + 9 \cdot 12 + 12 \cdot 16 + 3 \cdot 20 + 2 \cdot 24 + 2 \cdot 28) = \frac{1}{35} \cdot 552 \approx$$

$$\approx 15,77$$

$$c) M_0 = 14 + \frac{12-9}{(12-9)+(12-3)} \cdot 4 = 14 + \frac{3}{12} \cdot 4 = 15$$

$$M_0 = 15$$

$$d) N = 35$$

$$\frac{N}{2} = 17,5$$

$$M_e = 14 + \frac{17,5 - 14}{12} \cdot 4 = 14 + \frac{3,5}{12} \cdot 4 = 14 + 1,166666667 \approx 15,17$$